

From law of motion,

$$\vec{F}_R \propto -\vec{x}$$

$$\vec{F}_R = -kx$$

$$\frac{dp}{dt} = -kx$$

$$\Rightarrow \frac{d(mv)}{dt} = -kx$$

$$\Rightarrow m \frac{dv}{dt} = -kx$$

$$\Rightarrow m \frac{d}{dt} \left(\frac{dx}{dt} \right) = -kx$$

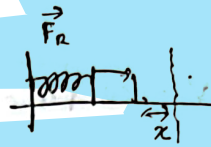
$$\Rightarrow m \frac{d^2x}{dt^2} = -kx$$

$$\Rightarrow \frac{d^2x}{dt^2} = -\frac{k}{m}x$$

$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

mass-spring satisfy SHM
/ differential equation of SHM

mass-spring



$$\frac{dp}{dt} = \vec{F}$$

x = displacement

$$p = mv \rightarrow \text{eq (asr)}$$

$$\omega^2 = \frac{k}{m}$$

$$v = \frac{dx}{dt}$$

Energy -

$$E_p = \frac{1}{2} kx^2$$

$$\text{let } x = A \cos \omega t$$

$$= \frac{1}{2} k A^2 \cos^2 \omega t$$

$$E_k = \frac{1}{2} mv^2$$

$$= \frac{1}{2} m A^2 \omega^2 \sin^2 \omega t$$

$$= \frac{1}{2} m A^2 \frac{k}{m} \sin^2 \omega t$$

$$= \frac{1}{2} k A^2 \sin^2 \omega t$$

$$\text{Total energy} = E_p + E_k$$

$$= \frac{1}{2} k A^2 \cos^2 \omega t + \frac{1}{2} k A^2 \sin^2 \omega t$$

$$= \frac{1}{2} k A^2 (\sin^2 \omega t + \cos^2 \omega t)$$

$$E_T = \frac{1}{2} k A^2 \quad (\text{Total energy doesn't depend on time})$$

Qn: Show that $v = \omega \sqrt{A^2 - x^2}$ / Derive the equation of v

$$E_T = E_P + E_K$$

$$\Rightarrow \frac{1}{2} k A^2 = \frac{1}{2} k x^2 + \frac{1}{2} m v^2$$

$$\Rightarrow \frac{1}{2} m v^2 = \frac{1}{2} k A^2 - \frac{1}{2} k x^2$$

$$\Rightarrow m v^2 = k A^2 - k x^2$$

$$\Rightarrow m v^2 = k (A^2 - x^2)$$

$$\Rightarrow m \cdot v^2 = \frac{k}{m} (A^2 - x^2)$$

$$\Rightarrow v = \sqrt{\frac{k}{m} (A^2 - x^2)}$$

$$\Rightarrow \boxed{v = \omega \cdot \sqrt{A^2 - x^2}} \rightarrow \text{no need time}$$

$$\therefore v = \omega \sqrt{A^2 - x^2}$$

proved

$$\boxed{v = -A\omega \sin \omega t} \rightarrow \text{Time that starts}$$



Q₂: Show that change of motion satisfy S.H.M



LC

We know,

$$Q = CV_c$$

$$\Rightarrow V_c = \frac{Q}{C}$$

$$V_L = L \frac{di}{dt}$$

V_L = inductor

V_c = voltage capacitor

Q = plate voltage

$$i = \frac{dQ}{dt}$$

now, $V_L + V_c = 0$ [Kanshof law]

$$\Rightarrow L \frac{di}{dt} + \frac{Q}{C} = 0$$

$$\Rightarrow L \frac{d}{dt} \left(\frac{dQ}{dt} \right) + \frac{Q}{C} = 0$$

$$\Rightarrow L \frac{d^2Q}{dt^2} + \frac{Q}{C} = 0$$

$$[\omega^2 = \frac{1}{LC}]$$

$$\Rightarrow \frac{d^2Q}{dt^2} + \frac{1}{LC} \cdot Q = 0$$

$$\Rightarrow \frac{d^2Q}{dt^2} + \omega^2 Q = 0$$

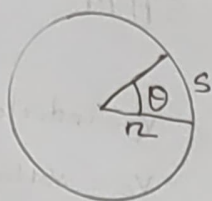
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$Q = Q_0 \cos \omega t$ is the most sweetable solution



Derivation Equation

Q1M



Qn: Show, simple pendulum satisfies the differential equation of SHM

$$s = r\theta$$

$$x = L\theta$$

$$\frac{dx}{dt} = L \frac{d\theta}{dt}$$

$$\frac{d^2x}{dt^2} = L \frac{d^2\theta}{dt^2}$$

$$a = L \frac{d^2\theta}{dt^2}$$

We know, $F = -mg \sin\theta$

$$\Rightarrow ma = -mg\theta$$

$$\Rightarrow a = -g\theta$$

$$\Rightarrow L \frac{d^2\theta}{dt^2} = -g\theta$$

$$\Rightarrow \frac{d^2\theta}{dt^2} = -\frac{g}{L}\theta$$

$$\Rightarrow \frac{d^2\theta}{dt^2} + \frac{g}{L}\theta = 0$$

$$\Rightarrow \frac{d^2\theta}{dt^2} + \omega^2\theta = 0$$

(showed)

[in angle $\omega^2 = \frac{g}{L}$]

[$\sin\theta = \theta$]



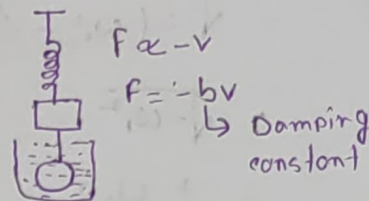
Q3! Derive the differential equation of DHM/
/ Damped H.M doesn't satisfy the SHM

DHM

We know,

$$F = ma = -kx - bv$$

Hooke's law
damping force



from, damping force,

$$F \propto -v$$

$$\Rightarrow ma = -kx - bv$$

$$\Rightarrow m \frac{d^2x}{dt^2} = -kx - bv$$

$$\Rightarrow \frac{d^2x}{dt^2} = -\frac{k}{m}x - \frac{b}{m} \frac{dx}{dt}$$

$$\Rightarrow \frac{d^2x}{dt^2} + \omega^2 x + \gamma \frac{dx}{dt} = 0$$

$$\Rightarrow \frac{d^2x}{dt^2} + \gamma \frac{dx}{dt} + \omega^2 x = 0 \quad \text{--- (1)}$$

From equation (1).

$$\text{let } x = Be^{pt}$$

$$\frac{dx}{dt} = Bpe^{pt}$$

$$\frac{d^2x}{dt^2} = Bp^2 e^{pt}$$

$$\therefore \frac{d^2x}{dt^2} + \gamma \frac{dx}{dt} + \omega^2 x = 0$$

$$Bp^2 e^{pt} + Bpe^{pt} + \omega^2 Be^{pt} = 0$$

$$\Rightarrow Be^{pt} (p^2 + \gamma p + \omega^2) = 0$$

$$\Rightarrow p^2 + \gamma p + \omega^2 = 0$$

$$\therefore p = \frac{-\gamma \pm \sqrt{\gamma^2 - 4\omega^2}}{2}$$

$$\therefore p = -\frac{\gamma}{2} \pm \sqrt{\frac{\gamma^2}{4} - \omega^2} \quad \text{--- (1)}$$

from equation (1)

$$\textcircled{i} \omega_0^2 > \frac{\gamma^2}{4} \quad (\text{Oscillatory})$$

$$\textcircled{ii} \omega_0^2 = \frac{\gamma^2}{4} \quad (\text{critical})$$

$$\textcircled{iii} \omega_0^2 < \frac{\gamma^2}{4} \quad (\text{Overdamped})$$

Qn: Find the solution for case (1, 2, 3) -

$$\textcircled{1} \omega^2 > \frac{\gamma^2}{4}$$

$$= \sqrt{\omega_0^2 - \frac{\gamma^2}{4}}$$

$$= \sqrt{-\omega_1^2}$$

$$= \pm \omega_1 j$$

$$\therefore P = -\frac{\gamma}{2} \pm \omega_1 j$$

from this,

$$P_1 = -\frac{\gamma}{2} + \omega_1 j$$

$$P_2 = -\frac{\gamma}{2} - \omega_1 j$$

now, $x = B e^{Pt}$

$$x = B_1 e^{P_1 t} + B_2 e^{P_2 t}$$

$$= B_1 e^{(-\frac{\gamma}{2} + j\omega_1)t} + B_2 e^{(-\frac{\gamma}{2} - j\omega_1)t}$$

$$= B_1 e^{-\frac{\gamma}{2}t} \cdot e^{j\omega_1 t} + B_2 e^{-\frac{\gamma}{2}t} \cdot e^{-j\omega_1 t}$$

$$= e^{-\frac{\gamma t}{2}} (B_1 e^{j\omega_1 t} + B_2 e^{-j\omega_1 t})$$

$$= A e^{-\frac{\gamma t}{2}} \cos(\omega_1 t + \phi)$$

$$\textcircled{2} \omega_0^2 = \frac{\gamma^2}{4}$$

$$P = -\frac{\gamma}{2} \pm \sqrt{\frac{\gamma^2}{4} - \omega_0^2}$$

$$= -\frac{\gamma}{2}$$

$$\therefore x = (A + Bt) e^{-\frac{\gamma t}{2}}$$

$$\omega_0^2 = \frac{\gamma^2}{4}$$

(iii)

$$\omega_0^2 < \frac{\gamma^2}{4}$$

$$x = B e^{Pt}$$

$$P = -\frac{\gamma}{2} \pm \sqrt{\frac{\gamma^2}{4} - \omega_0^2}$$

$$P_1 = -\frac{\gamma}{2} + \lambda$$

$$P_2 = -\frac{\gamma}{2} - \lambda$$

$$\gamma = \frac{b}{m}$$

$$\omega_0^2 = \frac{k}{m}$$

here, b, k, m all are constant so, its

calculation is also constant

$$\therefore x = B_1 e^{(-\frac{\gamma}{2} + \lambda)t} + B_2 e^{(-\frac{\gamma}{2} - \lambda)t}$$

Qn: Show RLC circuit satisfy DMM

RLC

We know,

$$V_L = L \frac{di}{dt}, \quad V_C = \frac{Q}{C}, \quad V = IR$$



now,

$$V_L + V_C + V = 0$$

$$\Rightarrow L \frac{di}{dt} + \frac{Q}{C} + IR = 0$$

$$\Rightarrow L \frac{d}{dt} \left(\frac{dQ}{dt} \right) + R \frac{dQ}{dt} + \frac{Q}{C} = 0$$

$$\Rightarrow \frac{d^2Q}{dt^2} + \frac{R}{L} \frac{dQ}{dt} + \frac{1}{LC} Q = 0$$

$$\Rightarrow \frac{d^2Q}{dt^2} + \gamma \frac{dQ}{dt} + \omega_0^2 Q = 0$$

showed for DMM

in RLC,

$$\gamma = \frac{R}{L}$$

$$\omega_0^2 = \frac{1}{LC}$$